

The Speed of Information Revelation and Eventual Price Quality in Markets with Insiders: Comparing Two Theories*

PETER BOSSAERTS¹, CARY FRYDMAN² and JOHN LEDYARD³

¹*HSS and CNS, California Institute of Technology*; ²*Marshall School of Business, University of Southern California*; ³*HSS, California Institute of Technology*

Abstract. Two theoretical literatures, one using Bayesian Nash equilibrium (BNE), and the other using noisy rational expectations equilibrium (NREE), both provide a foundation for understanding how private information is impounded into asset prices, yet some of their predictions are conflicting. Here, we compare for the first time, the two theories using data from carefully controlled laboratory asset markets. In the dynamics, we find strong evidence for BNE theory, although final prices support predictions of the NREE theory. Finally, we document that price volatility increases when information is being impounded in prices.

JEL Classification: G12, G14, C92

1. Introduction

What is the speed at which private information is incorporated into the market price of a security? Does the number of informed traders in an economy affect the price discovery process? These and other questions have been closely examined in two separate theoretical literatures: one using Bayesian Nash equilibrium (BNE) (Kyle, 1985; Holden and Subrahmanyam, 1992) and the other using noisy rational expectations equilibrium (NREE) (Grossman and Stiglitz, 1980; Hellwig, 1980; Diamond and Verrecchia, 1981; Brennan and Cao, 1996). Results from the parallel literatures show similarities, but a critical difference involves the relationship between the number of informed agents and the informational efficiency of the final market price.

*Comments from Craig Holden and Avanidhar Subrahmanyam on an earlier draft, as well as discussions with Charles Plott and Sera Linardi are gratefully acknowledged. The usual disclaimer applies. Research supported by NSF grants DRU-0527491 and SES 0616431.

On the one hand, BNE assumes there is a strategic trading environment, and this should induce competition among insiders, resulting in brisk trading and an equilibrium governed by eventual full information revelation. In such a strategic environment with imperfect competition, each trader has the ability to impact the price through his own individual trading. Hence, equilibrium in this environment requires that all traders respond optimally to each other, which results in traders competing any informational rents to zero, and thus full information revelation. A rational expectations equilibrium, on the other hand, is rooted in the assumption that traders engage in perfect competition, and as such, do not have the ability to impact the market price through their trading. A NREE may not entail fully revealing prices, as traders may not be able to fully invert prices to infer information due to exogenous noise in the environment. Although the NREE and the BNE are both richly developed theories that make precise predictions about the informational efficiency of prices, empirical tests of these predictions have yet to be as successful (Biais *et al.*, 2010). These tests are necessary if we are to have a better understanding of the predictive power of each theory to explain trading patterns and prices in markets with asymmetric information. An inherent difficulty in performing such tests is the inability of the researcher to identify who holds information at what time. To sidestep this difficulty, we design an experiment to test how well each theory can describe data from carefully controlled laboratory asset markets.

Our experimental treatments allow us to examine the relationship between eventual price quality and the proportion of traders in an economy that has private information. We vary the number of traders in our experimental asset market that receive private information regarding the liquidating dividend of a risky security, which allows us to quantify the amount of information that is revealed to the market throughout each trading period. Moreover, our markets are run in a continuous double auction institution that provides us with time series data to study statistical properties of prices and volatility that we find to be associated with the information structure in the economy. We find that both the final price quality and speed of information transmission are associated with the number of informed agents, empirical facts that are predicted by the theories of NREE and BNE, respectively.

Our experimental design follows in the footsteps of Plott and Sunder (1988), and hence, is informed by asset pricing theory. This contrasts with other recent work in experimental finance on the impact of insiders, notably that of Schnitzlein (2002), where the design is rooted in market microstructure theory, and hence, game theory. There, control of common knowledge is key (Aumann, 1976), in contrast with asset pricing theory, where mere knowledge of prices (and good forecasting of future price paths) is needed

(Radner, 1972). Nevertheless, we demonstrate here that BNE theory does make valid predictions when agents do not have common knowledge about the underlying structure of the economy. One of the contributions of our work, therefore, is to establish that the essence of the dynamics in a standard experimental asset pricing setting continues to be well captured by game-theoretic inspired models (Holden and Subrahmanyam, 1992), even when the assumption of common knowledge is not satisfied. As a result, we are the first to empirically establish, within a common setting, a link between market microstructure theory and asset pricing theory. The need to do so had first been articulated in O'Hara (2003). Our main result fulfills this need by tying together the two strands of theoretical literature, as our data support the dynamics of the BNE theory (Holden and Subrahmanyam, 1992), and the final market price predicted by the NREE theory (Grossman and Stiglitz, 1980).

2. Theory

Asset pricing theory under asymmetric information has relied extensively on two competing notions of equilibrium behavior: NREE and BNE. For the purpose of microstructure analysis, the critical difference between the two equilibrium concepts involves the impact that each agent can have on the price. In particular, the NREE model does not allow the individual agent to impact the price; agents are competitive and take the price function as given when formulating demands for securities. In contrast, BNE is used to model strategic situations where individual agents can be “large” and are able to impact the price function given their demand schedules. In this section, we highlight the relevant aspects of both theories, and we ultimately produce hypotheses that we test using our laboratory-generated data.

2.1 NOISY RATIONAL EXPECTATIONS EQUILIBRIUM

Under the NREE framework (Grossman and Stiglitz, 1980; Hellwig, 1980; Diamond and Verrecchia, 1981) prices partially reveal inside information to the rest of the market. In what follows, we focus on the theory of Grossman and Stiglitz (1980), as our experimental design examines information *revelation* where each insider is endowed with the same piece of information¹, as opposed to information *aggregation* where signals of the underlying state of the economy are dispersed (Hellwig, 1980; Diamond and Verrecchia, 1981).

¹ In the original Grossman and Stiglitz (1980) model, inside information is acquired endogenously at a fixed cost. In our design, insiders are exogenously endowed with information for simplicity.

Moreover, although there exist dynamic models of NREE, they will not apply in our setting because our experiment does not involve intermediate consumption, nor does it involve the dissemination of new information or noise over time (Grundy and McNichols, 1989; Wang, 1994; He and Wang, 1995; Brennan and Cao, 1996).

In the Grossman and Stiglitz (1980) model, uninformed traders infer inside information from the market price, which, in equilibrium, is determined by a convex combination of the demands of the uninformed and informed segments of the market. The equilibrium weight placed on the informed agents' demand increases with the proportion of informed agents in the market. This ultimately leads the price to be more responsive to the insiders' signal in a market with a higher proportion of informed agents. Hence, the theory predicts that *ceteris paribus*, markets containing a higher proportion of informed agents (with homogenous signals) will have a more informative price at all times, where we define *informativeness* by

$$\frac{1}{|p_t - \theta|},$$

and where p_t denotes the price at time t and θ denotes the signal the insider receives before the start of trading.

Because the NREE theory makes quantitative predictions about the equilibrium price efficiency as a function of the proportion of the economy that is informed, it is important to have sufficient exogenous variation in this variable to perform powerful statistical tests of the theory. We therefore vary the proportion of the informed agents in our markets over a wide parameter range: 0–0.75. This element of our design contrasts with Schnitzlein (2002) who uses markets with 0, 1, or 2 informed traders, because the focus of that paper is on the BNE theory. As we describe in the next section, BNE predicts that strategic competition among any $N > 1$ insiders should give rise to the same level of informational efficiency reflected in the final market price; hence, the design in Schnitzlein (2002) is not concerned with quantitatively assessing the role that the marginal insider plays in impounding information into asset prices. To put it another way, our design is aimed at testing different concentrations of oligopolistic competition, whereas Schnitzlein (2002) is concerned with testing between monopolistic and duopolistic competition.

2.2 BAYESIAN NASH EQUILIBRIUM

The BNE theory allows traders' orders to impact the market price, in contrast to the price-taking assumption of the NREE model. Hence, the

BNE theory gives rise to strategic interaction among traders where insiders may have an incentive to slowly reveal their private information over time. The seminal analysis of insider trading using BNE was done in Kyle (1985), where it is shown that an information monopolist will reveal his private information gradually over time to the market by submitting market orders. Holden and Subrahmanyam (1992) (HS) extend this to competition of long-lived private information between multiple insiders. This extension represents the type of imperfect competition that we envisaged in our experiments.

HS study a multi-period asset market with a single risky asset that pays a liquidating dividend, v , at the end of N periods. The dividend value is normally distributed and only the informed agents in the economy see the realized value of the liquidating dividend before the first trading period. Hence, there are only two types of agents, the informed and the uninformed, and each informed agent has identical information (a more general setup allowing for correlated signals across insiders is examined in Foster and Viswanathan (1996)). HS show that several market variables, including the informational efficiency of price, are a function of the number of insiders in the market. In particular, they find that as the number of informed agents in the economy increases, so too does the informational efficiency of the price, as measured by the conditional variance of the equilibrium market price in each trading period. See Figure 1, which is reproduced from HS. Intuitively, competition causes insiders to be more aggressive early on, for fear of others stealing their information rents. In the limit, as the number of insiders approaches infinity, all information is incorporated into the market price in the first trade.

However, although HS find that information is revealed at different speeds as a function of the number of insiders, they also conclude that the “informational advantage in financial markets is competed away extremely rapidly even in the case of just two informed traders” (Holden and Subrahmanyam, 1992, p. 257). By the last period of trade, all inside information is incorporated into the price. The predictions of HS contrast with those from Bertrand competition. Under Bertrand competition, from the moment there are two insiders with enough financial clout to potentially buy or sell the entire supply of securities at hand, all inside information is revealed instantaneously.

Prior experimental work (Schnitzlein, 2002) has shown that when the assumption in HS of common knowledge of the number of insiders in the market is violated, the dynamics of the HS framework are not achieved. However, the experimental design in Schnitzlein (2002) is of a very different nature from ours. Like the HS theory, it is heavily rooted in game theory,

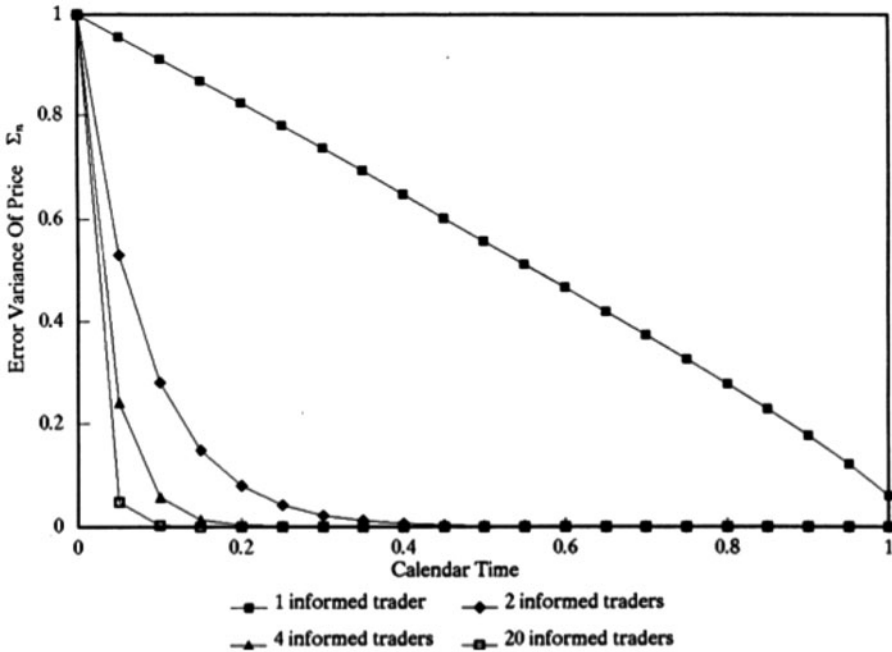


Figure 1. As the number of competing insiders increases, the price quality augments at a decreasing rate, and hence, the error variance in the price relative to the inside information decreases. *Source:* Reproduced from Holden and Subrahmanyam (1992).

and hence common knowledge is crucial. Our design is inspired by asset pricing theory, which builds on knowledge of prices and accurate (“rational”) expectations of future prices², and hence *does not invoke* common knowledge in order to arrive at equilibrium pricing. Interestingly, our experimental results imply that several of HS’ predictions are robust to the paucity of common knowledge³.

In our experimental markets, we exogenously manipulate the amount of private information in the economy by endowing a different proportion of 20 traders with private information across different periods. This simple

² Motives for trade in our experiment are explicitly based on risk aversion. Moreover, there is no aggregate risk in the economy, which means that asset pricing theory predicts prices will behave “as if” every subject is risk neutral. This allows us to interpret prices without concern for (unobservable) levels of individual risk aversion.

³ One potential reason for the different effect of common knowledge in our design compared with that of Schnitzlein (2002), is that in our design, the focus is on uncertainty over the number of insiders within an oligopoly information structure. In the Schnitzlein (2002) design, there is uncertainty over the *type* of information structure: monopoly, duopoly, or oligopoly.

treatment technique allows us to formulate testable hypotheses based on the previously discussed theories:

Hypothesis 1A (BNE)

Under the BNE theory, the speed at which the price converges to the underlying fundamental is positively correlated with the proportion of informed agents in the market.

Hypothesis 1B (BNE)

Under the BNE theory, although the speed of information transmission varies with the proportion of insiders in the market, all private information is eventually incorporated into the market price.

Hypothesis 2 (Bertrand)

Under Bertrand competition, we expect insiders to immediately compete away any informational profits, and the price will instantaneously reflect all inside information.

Hypothesis 3 (Noisy REE)

Following Conjecture 1 from Grossman and Stiglitz (1980, p. 394), “The more individuals who are informed, the more informative is the price system.” However, the equilibrium price is never fully revealing.

Because our experimental setting does not include intermediate consumption or the introduction of new information or noise over the course of a trading period, the dynamic NREE models do not apply in our setting. Hence, we refrain from making any predictions regarding the dynamics of pricing from the NREE model. It would be interesting to develop a model of dynamic NREE that corresponds to our experimental setting, but we leave that for future research.

2.3 PRICE VOLATILITY AND INFORMATION TRANSMISSION

Key to the theory of asset pricing under asymmetric information is the ability of uninformed traders to infer private information from the market price; that is, uninformed traders are assumed to be able to invert the price function, even if only imperfectly. Much of the theory on information transmission and revelation takes the ability of the uninformed to do this as given, and voluminous experimental research, starting with Plott and Sunder (1988), has repeatedly shown that even novice traders are quite capable of correctly reading information from prices (Lundholm, 1991; Ackert, 2008).

A recent neuroeconomic study (Bruguier, Quartz, and Bossaerts, 2010) shows that, during replay of markets with insiders, uninformed traders engage theory of mind, an innate human capacity to detect and track intentionality in their environment. Theory of mind is thought to rely on pattern recognition, but it is unknown what these patterns (statistical properties of price dynamics) are in the case of markets with insiders that allow the uninformed to track the information of the informed. Bruguier, Quartz, and Bossaerts (2010) provide preliminary evidence that the trading decisions of insiders generate excess volatility, which uninformed market participants can use to ascertain their presence. Likewise, in field data, price volatility is higher during trading than when markets are closed, a finding that is generally attributed to the presence of insiders, who of course have to rely on trade (and hence, open markets) to profit from their informational advantage (French and Roll, 1986)⁴.

These findings prompt us to conjecture the following about the link between insider information and volatility⁵:

Hypothesis 4

Volatility is highest at the beginning of the period when the information asymmetry is strongest, and it decreases over time as information is incorporated into the market price.

Our reasoning is that at any point in time, the price in a market with fewer insiders is less informative, and hence there is more information yet to be incorporated into the price. In summary, we hypothesize that markets with insiders can be characterized by the time series of volatility, and furthermore, this time series may have the power to reveal the proportion of market participants that is informed.

3. Experimental Design

To generate our laboratory data, we run a series of virtual stock markets using an anonymous, online platform, like Euronext and other electronic financial markets in the real world, which uses the continuous open book

⁴ See also Ederington and Lee (1993), who show that an increase in volatility is often a signal that private information is entering the market.

⁵ The equilibrium volatility is difficult to compute analytically under the HS theory. Furthermore, our conjectures about volatility are inherently dynamic, and hence are not predicted by the static NREE theory. It follows that our conjectures are not necessarily rooted in either the HS or NREE theory, but instead are meant as exploratory hypotheses about the mechanism through which the uninformed can infer information from statistical regularities in pricing.

exchange mechanism. The platform is open-source and is fully described online at <http://jmarkets.ssel.caltech.edu/>. A screenshot of the trading interface and instructions, which are read aloud to subjects while they follow along on the computer, are located at <http://clef.caltech.edu/exp/info/>.

A total of five *sessions* are run, each with twenty different subjects and each session consisting of thirteen independent *periods*⁶. The periods have a duration of 5 min, during which subjects are allowed to trade by submitting anonymous, but publicly displayed, orders.

Before trade starts in each period, subjects are endowed with notes, cash, and two risky securities, all of which expire at the end of the period. The two risky securities (stock X and stock Z) pay complimentary dividends, and the notes always pay 50 cents. Denote the dividend paid by stock X as v , so that stock Z pays $50-v$ cents. Endowments are heterogeneous across subjects, but the aggregate supply of the two risky securities is equal so that there is no aggregate risk. Theoretically, risk neutral pricing should obtain: prices should equal the expected payoff of the asset.

We appeal to risk (or ambiguity) aversion to induce trade. We allow trade only in stock X and the note, and *we close market Z for trading*. As such, participants endowed with more units of stock X than stock Z can obtain a balanced (risk-free) portfolio only by selling X, whereas participants endowed with more of stock Z than stock X can balance their portfolio only by buying stock X. Because there are an equal number of participants in both categories, and because there was no a priori reason to expect that subjects in either category are more risk or ambiguity averse than in the other, equal price pressure should obtain from the sell and buy side.

This design is a major departure from traditional experiments on asset pricing under asymmetric information, which either start everyone out with an equal endowment of a single risky security so that there is aggregate risk (e.g. the prediction markets of Arrow *et al.* (2008)), or which use differences in payoffs unknown to the participants (e.g. Plott and Sunder, 1988; Camerer and Weigelt, 1991). We prefer our approach because it induces trade without a priori imputing a (unknowable) bias in pricing due to the risk or ambiguity aversion that is inevitably present in experiments (Bossaerts and Plott, 2004; Bossaerts, Plott, and Zame, 2007; Bossaerts *et al.*, 2010).

In each session, two of the thirteen periods are designated as “control” periods, where no trader receives any information regarding the final liquidating dividend of stock X. The remaining eleven periods are designated

⁶ Hence, we use a between-subjects design so as to maximize exogenous variation in the number of insiders across the 65 periods.

“test” periods, where we exogenously manipulate the proportion of the market that is informed. There are exactly twenty subjects in each period, and the number of informed agents is manipulated between two and fifteen traders. Table I displays the number of insiders and the signal given to the insider for each of the sixty-five periods. In periods with insiders, all agents know that there are insiders. However, in some periods, only insiders know how many insiders there are, and in other periods, no traders know the number of insiders⁷.

Insiders receive a signal for the period that is drawn uniformly from an interval within 5 cents of the liquidating dividend of stock X: $\theta \in [v - 0.05, v + 0.05]$.⁸ All insiders in a period receive the same signal. It is important to note that because insiders are endowed with an imperfect signal of the liquidating dividend, they too should engage in trading to hedge endowment risk.

Short selling is allowed, and subjects are instructed that the liquidating value of their short position will be deducted from their earnings at the end of the period. Period earnings are equal to the sum of final cash holdings plus any dividends paid for units of stock and notes held (or subtracted for units of stock and notes sold short). We also impose a bankruptcy constraint that bans the submission of orders that, if executed, could lead to a negative portfolio value for at least one possible value of the liquidating dividend. For insiders, the bankruptcy constraint is evaluated only for those possible values of the liquidating dividend that are consistent with their signal (e.g. if their signal is 0.20, then the bankruptcy constraint is evaluated only for dividend values equal to 0.15, 0.16, ..., 0.25). Additionally, this bankruptcy constraint takes into account any open orders, so that subjects cannot place orders that can lead to bankruptcy conditional on open orders being executed.

Earnings are cumulative across periods, and final payouts include a \$10 show-up reward. All sessions are run at Caltech's Social Sciences Experimental Laboratory, drawing participants from a pool of mostly graduate and undergraduate students, and a few postdoctoral fellows and staff. Each session lasts just over 2 h, of which 1 h was devoted to instructions

⁷ This is the violation of common knowledge that we referred to above. Note that this feature of the design provides a strong test of the HS theory because if predictions from this theory are upheld without explicit common knowledge of the underlying structure of the economy, this provides evidence for the robustness of the HS theory to certain informational properties of the economy.

⁸ Signals are truncated to fall within the possible range of dividends. For example, if the liquidating dividend is 0.49, then signals are drawn uniformly from [0.44, 0.5].

Table I. Experimental design

Each row indicates a single period, ordered by number of insiders. Actual experimental sessions consisted of 13 periods each. *First column*: number of insiders. *Second column*: absolute difference between final market price and insider signal. *Third column*: insider signal (“x” indicates no signal).

Insiders	Difference	Signal	Insiders	Difference	Signal
0	0	×	6	0.04	0.05
0	0	×	6	0.13	0.49
0	0	×	6	0.13	0.46
0	0.01	×	10	0	0.09
0	0.01	×	10	0	0.12
0	0.01	×	10	0	0.41
0	0.01	×	10	0	0.08
0	0.02	×	10	0	0.08
0	0.02	×	10	0	0.4
0	0.03	×	10	0.01	0.15
2	0.01	0.14	10	0.01	0.14
2	0.02	0.31	10	0.01	0.12
2	0.03	0.31	10	0.01	0.06
2	0.04	0.13	10	0.01	0.45
2	0.05	0.19	10	0.01	0.06
2	0.07	0.4	10	0.01	0.45
2	0.08	0.45	10	0.02	0.41
2	0.08	0.45	10	0.02	0.28
2	0.09	0.42	10	0.02	0.31
2	0.09	0.48	10	0.02	0.4
2	0.11	0.14	10	0.02	0.31
2	0.12	0.37	10	0.03	0.06
2	0.15	0.49	10	0.06	0.46
2	0.21	0.49	14	0.01	0.41
2	0.23	0.49	14	0.01	0.15
6	0	0.12	14	0.01	0.05
6	0.01	0.12	14	0.01	0.41
6	0.02	0.14	14	0.04	0.46
6	0.02	0.46	15	0	0.05
6	0.03	0.37	15	0	0.05
6	0.03	0.07	15	0.01	0.19
6	0.04	0.37	15	0.02	0.01
			15	0.02	0.46

(continued)

and training in using the trading interface. Instructions are located at <http://clef.caltech.edu/exp/info/>. Total earnings vary from \$35 to \$75 for a session.

Note that we do not explicitly impose all assumptions of the Kyle (1985) and HS models. In particular, we do not model noise traders that submit normally distributed orders, nor do we model the market maker who sets the price equal to the expectation of the liquidating dividend conditional on the price history. Moreover, we allow uninformed liquidity traders to enter orders at their own

discretion, which gives rise to the possibility that informed traders and liquidity traders exhibit trading activity that is concentrated during specific time intervals (Admati and Pfleiderer, 1988). We abstract from several of these assumptions, as our aim is to test whether theories invoking BNE or NREE have predictive power in a standard trading mechanism, namely, the continuous double auction. They should, if the theories capture the essence of the situation at hand.

4. Results

4.1 EVENTUAL PRICE QUALITY

In this section we report results from all sixty-five periods (thirteen periods in five sessions). Periods are differentiated by the number of insiders and the signal that the insiders receive, if applicable. Terminal market prices, along with the signal and number of insiders for each period are shown in Table I. The average (standard error) of the difference between the insider's signal and the final market price is \$0.035 (\$0.004), indicating that the final market price is statistically different from the insider signal ($p < 0.001$). This is consistent with results from Schnitzlein (2002) who finds that the mean price error in the final 20 s of trading is different from the insider's signal, but this result is inconsistent with BNE theory as outlined in Hypothesis 1B.

Figure 2 aggregates the sessions by number of insiders, and displays a measure of the mean price error at different points within the period⁹. The mean price error is defined as the absolute value of the difference in the market price p_t and the insider signal θ , $|p_t - \theta|$, averaged across all periods with the same number of insiders. Our first main result is that, unconditional on the number of insiders, price quality monotonically increases over time. This immediately rejects Hypothesis 2, the Bertrand Hypothesis, which claims that prices should immediately reflect the insider information.

As a control, we run ten periods with no insiders, where we define the signal as the unconditional expected dividend, \$0.25. Note that the control periods are similar to the treatment where all traders are informed (which we did not run), in the sense that all traders have symmetric information¹⁰; this is why the figure suggests a similarity between the treatment with fifteen

⁹ For all subsequent analyses, we pool data from the treatments with fourteen and fifteen insiders because there are no significant differences between these treatments, and it increases statistical power for our tests across the remaining treatments.

¹⁰ While all traders have symmetric information in the control periods and in the treatment where all traders are informed, the difference is in the *precision* of the information. In the control period, there is little precision because conditional on the subject's information set, there is uniform probability that the liquidating dividend is in the range of $[0, 0.5]$. In the

insiders (where so many traders are informed that their information is inevitably revealed in prices) and the control sessions with zero insiders. Because we set out to examine price patterns as a function of the number of insiders, all subsequent analyses use only periods where there are at least two insiders.

Our second main result is that the market price quality at the end of a period is correlated with the number of informed agents in the economy. This is clear from Figure 2, which shows that the mean price error at the end of the period decreases monotonically with the number of insiders. We estimate an OLS regression of the price error on the number of insiders and the informativeness of the signal using the fifty-five periods where at least two traders are informed¹¹:

$$|p_{\text{final}} - \theta| = \alpha + \beta_1 \text{ Insiders} + \beta_2 |0.25 - \theta| + \varepsilon$$

We find that $\beta_1 = -0.007$, which is significantly negative ($p < 0.001$); this indicates that increasing the proportion of the market that is informed by 5% (1 trader out of a total of 20) leads to an increase in eventual price quality of 0.007 cents, or a 1.4% increase in price efficiency (relative to the total price range of 0–50 cents). We also find that $\beta_2 = 0.398$ which is significantly positive ($p < 0.001$). This indicates that a more informative signal (one that is drawn near the extremes of the range $[0, 0.5]$) also leads to an increase in eventual price quality. One potential explanation for this result is that it may be easier for uninformed traders to learn from the market price when the signal is “far” from their ex-ante expectation of the liquidating dividend: \$0.25. Moreover, using only these two regressors, we find the R-squared from the regression to be 0.55, suggesting that this simple bivariate linear model has substantial predictive power.

This result supports the NREE hypothesis (Hypothesis 3), originally put forth in Grossman and Stiglitz (1980). Conversely, this result rejects the notion that competition between insiders should drive prices (immediately) toward the fully revealing equilibrium via Bertrand pricing (Hypothesis 2). Because the final market price, on average, does not reach the fully revealing price, we also reject Hypothesis 1B, which states that all information will eventually be reflected in prices, regardless of the number of insiders.

An important question given our results is, what is the source of the end-of-period mispricing? One possibility is that the mispricing is due to the stochastic aggregate supply of risky asset shares, which is precisely the source of the mispricing in the original Grossman and Stiglitz (1980) model.

treatment where all traders are informed, there is much greater precision as there is uniform probability that the liquidating dividend is in the range $[\theta - 0.05, \theta + 0.05]$

¹¹ Each period was treated as an independent observation.

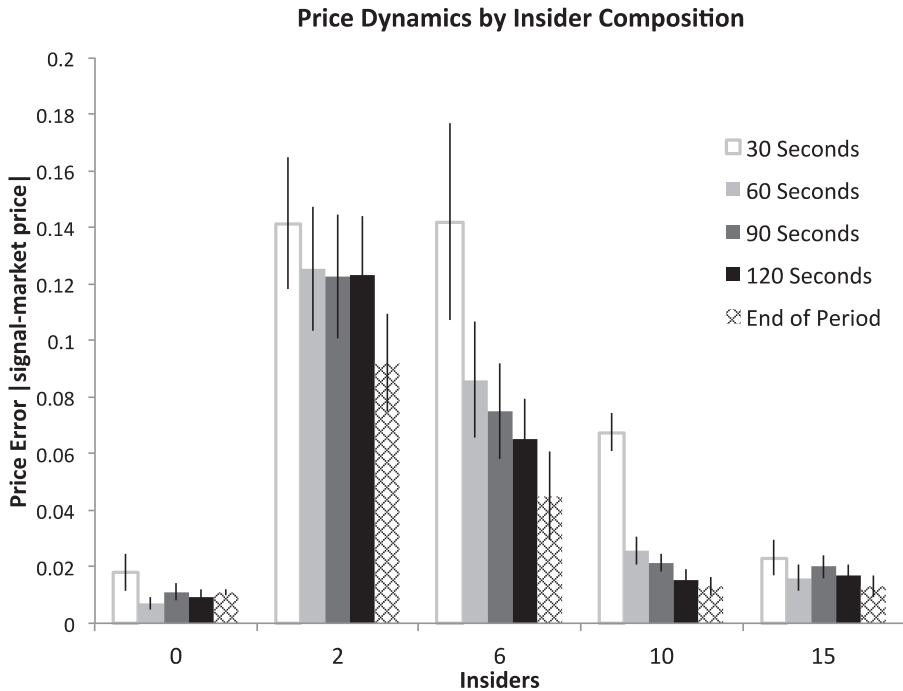


Figure 2. Price quality (absolute difference between market price and insider signal) after 30, 60, 90, 120 s, and at the end of a period (300 s), stratified by number of insiders (out of 20 participants). Averages are taken across periods with the same number of insiders (different insider signals). Vertical bars depict interval around average of length equal to two standard errors.

While the aggregate supply of shares of the risky stock in our experiment is a fixed parameter at the beginning of each period, set by the experimenter, it is unknown to every subject. Hence, in the tradition of Bayesian analysis, within each subject's information set, the aggregate supply is in fact a random variable, and the mispricing may be driven by this stochasticity.

Another possibility is that the mispricing is driven by trading based on subject error. Because the size of the mispricing is highly correlated with the number of informed traders, it is likely the case that if the source of mispricing is due to subject error, it is on behalf of the informed trader subjects, who effectively “leave money on the table” at the end of the period. This is consistent with past experimental work showing that subjects may not always compete prices down to cost in a Bertrand setting, which can be driven by a small probability of some subjects setting prices irrationally (Dufwenberg and Gneezy, 2000). While we speculate that the mispricing is driven by either stochastic aggregate supply or

subject error, our data do not allow us to identify which of the two, if either, are driving the wedge between final prices and the insider signal¹².

We conclude that the noisy REE theory explains final price quality of our asset markets with asymmetric information better than either the BNE or Bertrand competition theories.

4.2 SPEED OF INFORMATION REVELATION

Our third main result concerns the speed at which private information is transmitted to the market. Figure 2 shows a significant difference in the amount of information transmitted to the market within 30 s, between the treatments of fifteen insiders and two insiders. With two insiders, the market has yet to incorporate nearly any information in the price: the market price is 14 cents away from the insider signal. In contrast, with fifteen insiders, nearly all information is reflected in the price: the market price is a mere 2 cents away from the insider signal.

This suggests that, with more insiders, the price adjustment process is more rapid. Indeed, the speed of price adjustment within the first 30 s (measured as the difference in the market price and the conditional expected dividend, at 30 s) is correlated with the number of insiders ($p < 0.001$) as theorized by HS (see Figure 1, reproduced from their article), and encapsulated in our Hypothesis 1A.

Although our data qualitatively support the dynamics of HS, the speed with which information is revealed through trading is substantially slower in our experiments than in the theory. Figure 1 suggests that nearly all information should be incorporated in the price within the first one-third of the trading period. Our data (Figure 2) do not support this: with two or six insiders, price quality improves significantly during the latter two-third of the trading period ($p < 0.001$, under the null that price quality improvement is absent; Tobit model corrected for left-censoring at 0).

We also cast doubt on the prediction of the HS theory that the prices should converge to the fundamental by the end of the period, independent of the number of insiders (Hypothesis 1B), as the eventual price quality in our experiments is significantly correlated with the number of insiders ($p < 0.001$). Altogether, while the data support the hypothesis that the speed of information revelation is a function of the number of informed agents in the market, eventual pricing is not as “informationally efficient” as predicted in HS.

¹² Future work would be able to test the noisy aggregate supply hypothesis by exogenously varying the uncertainty of the aggregate supply from the subject's point of view.

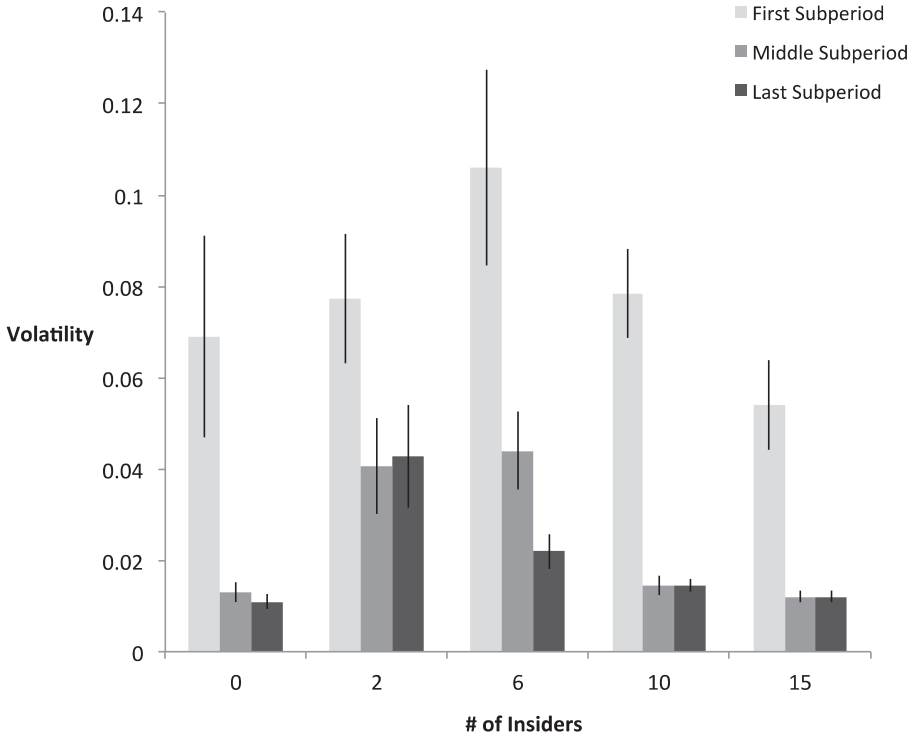


Figure 3. Average volatility (measured as transaction price range) in three subperiods of 100s (First, Middle, and Last), stratified by number of insiders. Averages are taken across periods with the same number of insiders (different insider signals). Vertical bars depict interval around average of length equal to two standard errors.

4.3 VOLATILITY

Information aggregation experiments have repeatedly confirmed the ability of uninformed traders, even novices, to readily detect insider trading. Building on preliminary evidence from Bruguier, Quartz, and Bossaerts (2010), we conjecture that patterns in volatility reveal not only the presence, but also the number, of insiders (Hypothesis 4).

We divide each period into several subperiods, and use the range of transaction prices as our measure of volatility¹³. Figures 3 and 4 document the

¹³ We also studied the dynamics of volatility when measured as the standard deviation of the error term in second-by-second regressions of transaction prices on insider signal across all sessions with a given number of insiders. This produces qualitatively the same outcomes. The graphs can be obtained upon request.

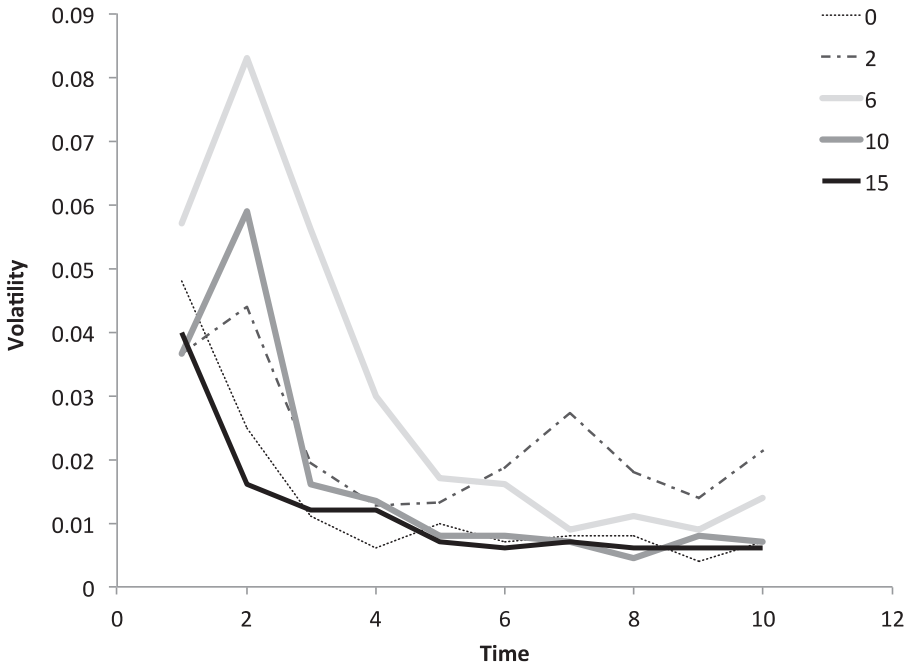


Figure 4. Average volatility (measured as transaction price range) per subperiod of 30 s, stratified by number of insiders (0, 2, 6, 10, and 15, out of 20 participants).

relationship between volatility and the number of insiders, when partitioning each period into three and ten equal subperiods, respectively.

Figure 3 shows that all markets exhibit heteroskedasticity. In the first subperiod, volatility is highest in markets with six insiders. In the final subperiod, volatility is highest in markets with two insiders. This suggests that, with two insiders, substantial insider trading may still be taking place near the end of the trading period. Overall, volatility is indeed negatively correlated with the number of insiders in the final subperiod ($p = 0.001$). This is consistent with the results in Figure 2 on the speed of information revelation, which show that markets with more insiders reveal information earlier. We also find that volatility in the middle subperiod is negatively correlated with the number of insiders ($p = 0.001$).

Figure 4 illuminates the fact that volatility in asset markets with two insiders *increases* midway through the trading period, whereas markets with more insiders generally see monotonically decreasing volatility after the first 60 s of trading. It is also interesting to note the similarity in the pattern of volatility between markets with no insiders and markets with fifteen insiders (when nearly all agents were informed).

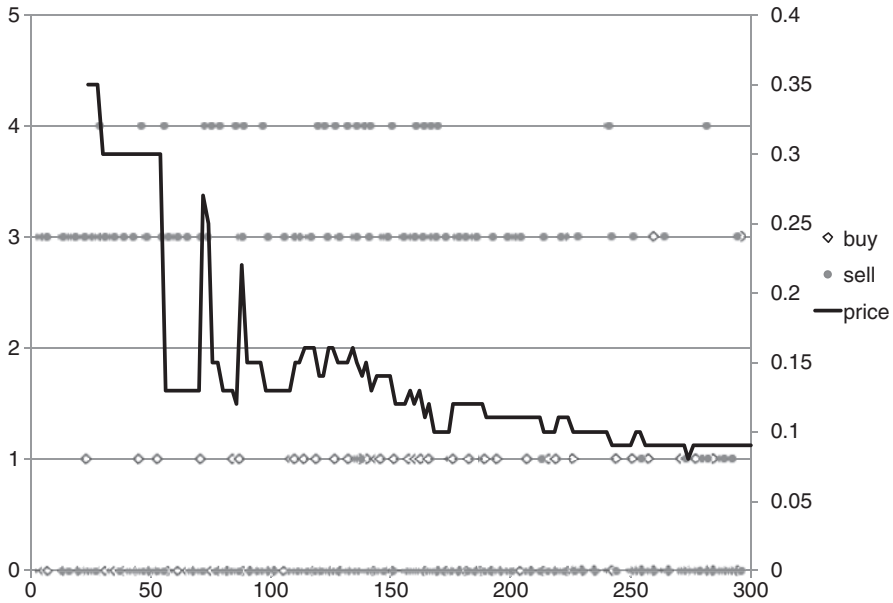


Figure 5. Transaction prices (solid black line; right scale) in one illustrative period with six insiders whose signal of the final dividend equals \$0.05. Timing of order flow, per identity of the order initiator (left scale), as follows: (0) uninformed limit order arrivals; (1) uninformed market order arrivals; (3) informed limit order arrivals; (4) informed market order arrivals.

4.4 FURTHER DETAILS

Although the main focus of this article is on price dynamics and informational efficiency of prices, our experiments produce a rich data set that can be used to examine other aspects of market microstructure theory. For example, Figure 5 shows the evolution of market and limit orders in an illustrative period with six insiders, stratified by submitter type (insider/uninformed). During this period, prices gradually decline to 9 cents, close to the insider signal of 5 cents.

Strikingly, insiders mostly submit limit (sell) orders (Bloomfield, O'Hara, and Saar, 2005). As such, insiders are actually providing liquidity, although one-sided. This contrasts with many formal models, including HS, where insiders are *assumed* to submit market orders, and where market makers provide liquidity. The ability of HS to shed light on our experimental data thus shows that the insights of the model do not depend on artificial order type restrictions imposed on market participants. In situations where insiders have the option to submit limit orders as well, thus providing

liquidity, some of the main conclusions evidently continue to obtain. Also note that uninformed and informed market order flow seem to be concentrated in specific time intervals, consistent with results in Admati and Pfleiderer (1988).

We are not the first to observe that insiders use limit orders. Our observation is in line with Barner, Feri, and Plott (2005), who also note that insiders tender limit order contracts. Because the focus in this article is on price dynamics and price quality, we leave a study of the microstructure details for future research.

5. Conclusion

There are two classes of models that make predictions about price quality in markets with insiders. One comes out of the tradition of competitive analysis and general equilibrium theory, and is known as NREE. The second one emerges from a game-theoretic analysis of market microstructure, and uses the concept of BNE. The two classes do not always make the same predictions. For example, the NREE theory predicts that prices will not fully reveal all inside information in the economy, but instead the price will be masked by noise in proportion to the number of informed agents in the market. The BNE theory predicts that, by the end of trading, competition for informational rents will drive the price to fully reveal all private information to the market.

In this article, we discuss results from a series of markets experiments where we exogenously vary the number of insiders over a wide range of parameters in order to examine its effect on the informational efficiency of prices. We discover that NREE best predicts the end-of-period price quality. Specifically, we find that price efficiency increases with the proportion of traders that hold inside information. This stands in contrast to the prediction that BNE makes about the (static) final market price efficiency: final market prices should be fully efficient *unconditional* on the proportion of the economy that is informed. On the other hand, the BNE theory quite effectively captures features of price dynamics in our experimental data. In particular, the speed with which information is revealed in prices depends on the number of insiders, in accordance with HS. Our experiments also provide positive results in light of Schnitzlein (2002), since we find that even in a far more strategically complex environment than the game-theoretic framework in HS, the dynamics of the HS theory emerge, while the NREE theory is able to provide an explanation for the fact that not all informational rents are captured at the end of trading. This finding allows us to uncover an

empirical connection between the literature on market microstructure and asset pricing (O'Hara, 2003).

One potential reason why we do not see fully revealing final pricing as predicted by the HS theory is lack of common knowledge about the number of insiders. Although lack of common knowledge does not seem to affect the predictions of the dynamics of HS, it is possible that there may be an effect during the final trading round. As we allude to above, there are other assumptions from the HS theory that we do not explicitly impose in our design. For example, we do not model noise traders as submitting normally distributed orders at each round. The insider's uncertainty about the uninformed traders' strategies may then affect the insider's equilibrium trading strategy. Another possible source of the (only) partially revealing prices is subject error. It may prove fruitful for theorists to model the environment in HS with boundedly rational insiders, so we can rigorously understand whether subject error can in fact induce this mispricing.

We also discover that insider trading generates heteroskedasticity, which itself is a function of the number of insiders in the economy. Consistent with a conjecture in Bruguier, Quartz, and Bossaerts (2010), uninformed traders therefore should be able to detect not only the presence, but also the number, of insiders from volatility patterns. Our findings provide support for the standard interpretation of heteroskedasticity in field data, namely, that it reflects incorporation of private information into pricing (French and Roll, 1986; Ederington and Lee, 1993). Moreover, the connection we uncover between the number of insiders and the level of price volatility suggests an open avenue for future research on using measures of volatility to detect insider trading.

Our experimental design departs from the standard approach in experimental economics. In particular, we are careful to ensure that participants always have a reason to trade, even if no insiders are present. We avoid aggregate risk in order to obtain pricing predictions unaffected by risk or ambiguity aversion. By closing a market in the complementary asset, we assure that downward price pressure from traders eager to sell to avoid risk or ambiguity, is offset exactly by upward price pressure from traders wanting to buy to avoid (the same) risk or ambiguity. It should be added that we never observe "information mirages" (whereby market prices seem to reflect precise insider information, but in fact get it completely wrong (Camerer and Weigelt, 1991; Barner, Feri, and Plott, 2005)). It is an open question whether our design—conscious of ambiguity or risk aversion, aggregate risk, and deliberately avoiding no-trade theorems while staying within standard asset pricing theory—explains the absence of information mirages.

Our experiments generate a wealth of data that can be used in the future to study more detailed aspects of the microstructure of markets. This data can shed light on, for example, the trading strategies of insiders when they can choose freely whether to submit limit or market orders. In this respect, we discover that insiders predominantly trade through limit orders, and hence, provide liquidity (albeit one-sided), corroborating earlier evidence (Barner, Feri, and Plott, 2005), but standing in contrast to standard restrictions imposed on insiders in theoretical models, including HS.

References

- Ackert, L. F. (2008) What affects the market's ability to adjust for optimistic forecast bias? Evidence from experimental asset markets, *Journal of Economic Behavior and Organization* **66**, 358.
- Admati, A. R. and Pfleiderer, P. (1988) A theory of intraday patterns: Volume and price variability, *Review of Financial Studies* **1**, 3–40.
- Arrow, K. J., Forsythe, R., Gorham, M., Hahn, R., Hanson, R., Ledyard, J. O., Levmore, S., Litan, R., Milgrom, P., Nelson, F. D., Neumann, G. R., Ottaviani, M., Schelling, T. C., Shiller, R. J., Smith, V. L., Snowberg, E., Sunstein, C. R., Tetlock, P. C., Tetlock, P. E., Varian, H. R., Wolfers, J., and Zitzewitz, E. (2008) The promise of prediction markets, *Science* **320**, 877–878.
- Aumann, R. J. (1976) Agreeing to disagree, *The Annals of Statistics* **4**, 1236–1239.
- Barner, M., Feri, F., and Plott, C. R. (2005) On the microstructure of price determination and information aggregation with sequential and asymmetric information arrival in an experimental asset market, *Annals of Finance* **1**, 73–107.
- Biais, B., Bossaerts, P., and Spatt, C. S. (2010) Equilibrium asset pricing and portfolio choice under asymmetric information, *Review of Financial Studies* **23**, 1503–1543.
- Bloomfield, R., O'Hara, M., and Saar, G. (2005) The “make or take” decision in an electronic market: Evidence on the evolution of liquidity, *Journal of Financial Economics* **75**, 165–199.
- Bossaerts, P. and Plott, C. (2004) Basic principles of asset pricing theory: Evidence from large-scale experimental financial markets, *Review of Finance* **8**, 135–169.
- Bossaerts, P., Plott, C. R., and Zame, W. (2007) Prices and portfolio choices in financial markets: Theory, econometrics, experiments, *Econometrica* **75**, 993–1038.
- Bossaerts, P., Ghirardato, P., Guarnaschelli, S., and Zame, W. R. (2010) Ambiguity in asset markets: Theory and experiment, *Review of Financial Studies* **23**, 1325–1359.
- Brennan, M. J. and Cao, H. H. (1996) Information, trade, and derivative securities, *Review of Financial Studies* **9**, 163–208.
- Bruguier, A. J., Quartz, S. R., and Bossaerts, P. (2010) Exploring the nature of “trader intuition”, *Journal of Finance* **65**, 1703–1723.
- Camerer, C. and Weigelt, K. (1991) Information mirages in experimental asset markets, *The Journal of Business* **64**, 463–493.
- Diamond, D. W. and Verrecchia, R. E. (1981) Information aggregation in a noisy rational expectations economy, *Journal of Financial Economics* **9**, 221–235.
- Dufwenberg, M. and Gneezy, U. (2000) Price competition and market concentration: An experimental study, *International Journal of Industrial Organization* **18**, 7–22.

- Ederington, L. H. and Lee, J. H. (1993) How markets process information: News releases and volatility, *The Journal of Finance* **48**, 1161–1191.
- Foster, F. D. and Viswanathan, S. (1996) Strategic trading when agents forecast the forecasts of others, *The Journal of Finance* **51**, 1437–1478.
- French, K. and Roll, R. (1986) Stock return variances: The arrival of information and the reaction of traders, *Journal of Financial Economics* **17**, 5–26.
- Grossman, S. J. and Stiglitz, J. E. (1980) On the impossibility of informationally efficient markets, *The American Economic Review* **70**, 393–408.
- Grundy, B. D. and McNichols, M. (1989) Trade and the revelation of information through prices and direct disclosure, *Review of Financial Studies* **2**, 495–526.
- He, H. and Wang, J. (1995) Differential informational and dynamic behavior of stock trading volume, *Review of Financial Studies* **8**, 919–972.
- Hellwig, M. F. (1980) On the aggregation of information in competitive markets, *Journal of Economic Theory* **22**, 477–498.
- Holden, C. W. and Subrahmanyam, A. (1992) Long-lived private information and imperfect competition, *The Journal of Finance* **47**, 247–270.
- Kyle, A. S. (1985) Continuous auctions and insider trading, *Econometrica* **53**, 1315–1335.
- Lundholm, R. J. (1991) What affects the efficiency of a market? Some answers from the laboratory, *The Accounting Review* **66**, 486–515.
- O'Hara, M. (2003) Presidential address: Liquidity and price discovery, *The Journal of Finance* **58**, 1335–1354.
- Plott, C. R. and Sunder, S. (1988) Rational expectations and the aggregation of diverse information in laboratory security markets, *Econometrica* **56**, 1085–1118.
- Radner, R. (1972) Existence of equilibrium of plans, prices, and price expectations in a sequence of markets, *Econometrica* **40**, 289–303.
- Schnitzlein, C. R. (2002) Price formation and market quality when the number and presence of insiders is unknown, *Review of Financial Studies* **15**, 1077–1109.
- Wang, J. (1994) A model of competitive stock trading volume, *Journal of Political Economy* **102**, 127–168.